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Top Skills

Battery Intelligence
Feature Engineering
Causal Inference

Honors-Awards

National Scholarship
Outstanding Graduate
Jiangsu Province Outstanding Student

Publications

Crystal growth, transport, and magnetic properties of antiferromagnetic semimetal $\text{Ni}_3\text{In}_2\text{Se}_2$

First-principles studies on the crystal structure and electronic properties of two-dimensional RuO_2 and OsO_2

Mode-selective anharmonicity induced by lone-pair electrons in layered oxyselenides

Patents

Analysis System and Method Based on First-Principles and Quantum Oscillation Testing

GUANZHANG LIU

Data Scientist | Battery Intelligence · Applied ML | Li Auto ·
ByteDance | Open to London Opportunities
London Area, United Kingdom

Summary

I'm a data scientist at Li Auto's Battery Intelligence department in Beijing, focused on applying machine learning and statistical modeling to large-scale EV battery data. My work spans the full data science pipeline — from designing feature systems over fleet-scale telemetry to building and deploying risk ranking and degradation models that directly inform product and after-sales decisions. Recent projects include a thermal risk identification system covering 1M+ vehicles under extreme class imbalance, and a quasi-causal degradation analysis projecting tens of millions of RMB in warranty savings for a single vehicle model. Prior to Li Auto, I spent four months at ByteDance as a data analyst on Douyin's market strategy team, building an IP popularity evaluation model that reached ~60% internal dashboard adoption and directly supported content investment decisions. I hold a Master's in Computational Materials Science from Nanjing University, graduating in the top 2% of my cohort. My academic background built strong foundations in mathematical modeling, statistical inference, and scientific computing — skills I've since applied across both internet and industrial AI settings. I work best in collaborative, intellectually open environments — the kind where ideas get pressure-tested in conversation before becoming decisions. I genuinely enjoy the back-and-forth of figuring out hard problems with people from different backgrounds, and I'm always looking to grow through the process. Currently open to data scientist roles in London.

Experience

Li Auto
Data Scientist
July 2024 - Present (2 years)
Beijing, China

| Applied quasi-causal inference to Li Auto's 1M+ fleet to identify and isolate the true drivers of SOH degradation from confounding variables — quantifying

that targeted usage strategy adjustments can extend battery life by ~30%; evolved the methodology into an AI-powered self-service analytics platform deployed across multiple vehicle models, projecting 20k+ RMB in warranty savings for a single vehicle model.

| Built an automated, end-to-end thermal runaway risk identification system for Li Auto's 1M+ vehicle fleet — reframing extreme class imbalance as a white-box risk-ranking problem; productized into a scalable, cross-platform ML scoring pipeline with per-vehicle explainable outputs across 5 battery platforms, enabling targeted after-sales inspection prioritisation.

| Designed and productized a battery behaviour feature engineering system onto an internal ML platform — custom event definitions, multi-operator aggregations, and cross-platform normalisation across 5 battery platforms; consolidated fragmented development workflows and improved feature iteration efficiency by ~85%.

| Designed a hybrid NLP pipeline for battery quality work-order classification — combining rule-based text matching, TF-IDF vectorisation, and Random Forest across 30+ fault categories; achieved ~90% accuracy on 15k+ samples, shifting analysts from full-sample labelling to confidence-guided review of hard cases, and establishing a reusable NLP foundation for downstream solution recommendation and quality monitoring.

| Delivered a portfolio of battery intelligence analytics spanning BMS thermal strategy validation, real-world user behaviour profiling, and life test optimisation across 3+ vehicle platforms — building systematic, end-to-end pipelines from time-series event detection to fleet-level statistical analysis; directly shaping R&D strategy and product validation standards.

ByteDance

Data Analytics

May 2023 - September 2023 (5 months)

Beijing, China

· Built a multi-source IP content popularity scoring system for Douyin's (Chinese version of TikTok) market strategy team — integrated consumption, dissemination, and search signals across 5+ internal and third-party data platforms, designed ML-enhanced pipeline monitoring to resolve upstream data instabilities, and automated previously manual data workflows; achieved

~60% dashboard adoption and validated ~40% growth in consumption metrics following targeted IP promotional campaigns.

- Delivered content strategy analytics for Qishui Music (Chinese music app) — redesigned music hotness metrics to fix time-comparability and data granularity flaws, built custom content vertical classification frameworks, and diagnosed new-user conversion rate decline through funnel analysis and KOL audience saturation modeling; findings directly shaped content selection strategy and marketing resource reallocation decisions.

Dongwu Securities Co.

Equity Research Analyst

December 2021 - May 2022 (6 months)

Nanjing

- Produced an in-depth equity research report on Harmonic Drive Systems, conducting multi-dimensional analysis across company fundamentals, competitive positioning, and market dynamics — built a mathematical model to forecast stock price trends and delivered short, medium, and long-term investment strategy recommendations.

Education

Nanjing University

Master of Science, Computational Materials Science · (September 2021 - June 2024)

Nanjing University

Bachelor of Science, Materials Science & Engineering · (September 2017 - July 2021)